THE ROW SUMS OF THE INVERSE OF THE BAND TRIANGLE $1/(3n+1)$, $k \le n \le 2k$, ARE THE PERIOD-DOUBLING SEQUENCE MODULO 2 (OEIS A127752)

MATH0

Abstract. Let A be the infinite lower-triangular matrix with entries $A(n,k) = 1/(3n+1)$ for $k \le n \le 2k$ and $A(n,k) = 0$ otherwise, and let $a(n)$ be the n-th row sum of A^{-1} (OEIS A127752). We prove Paul Barry's 2007 conjecture that $a(n) \bmod 2$ is the first Feigenbaum symbolic (period-doubling) sequence A035263, the fixed point of the morphism $0 \mapsto 11$, $1 \mapsto 10$; precisely, $a(n) \bmod 2 = \text{A035263}(n+1)$ for all $n \ge 0$. The proof derives $a(0) = 1$, $a(2m+2) = 3$ and $a(2m+1) = a(m) + 3$ from the system $A\,a = \mathbf{1}$ and compares this with the self-similarity of the morphic word. To our knowledge this is a new result: the conjecture is still marked as open in the OEIS, where it had been checked for the first 2048 terms. The proof is verified in Lean 4 against the frozen, fidelity-reviewed statement that math0 formalized from the OEIS entry, whose definitions are proved to agree with the matrix-inverse definition and are checked by the kernel against the listed terms of both sequences.

1. Introduction

Let $A = (A(n,k))_{n,k\ge 0}$ be the number triangle

$$A(n,k) = \begin{cases} \dfrac{1}{3n+1}, & k \le n \le 2k, \\ 0, & \text{otherwise.} \end{cases}$$

Row n of A has the entry $1/(3n+1)$ in the columns $\lceil n/2 \rceil \le k \le n$. Since A is lower triangular with nonzero diagonal entries $A(n,n) = 1/(3n+1)$, it has a unique lower-triangular inverse A^{-1}, and its leading $N \times N$ block is the inverse of the leading $N \times N$ block of A. The inverse is the triangle A127751 [9], and its row sums

$$a(n) = \sum_{k=0}^{n} A^{-1}(n,k) \qquad (n \ge 0)$$

form the sequence A127752 [10]: $1, 4, 3, 7, 3, 6, 3, 10, 3, 6, 3, 9, 3, 6, 3, 13, \ldots$ Both entries were submitted by Paul Barry on January 28, 2007. The entry A127752 contains the comment

```
a(n) mod 2 is first Feigenbaum symbolic sequence A035263 (conjecture).
```

Here A035263 [5] is the trajectory of 1 under the morphism $0 \mapsto 11$, $1 \mapsto 10$, also known as the period-doubling sequence: $1, 0, 1, 1, 1, 0, 1, 0, 1, 0, 1, 1, \ldots$ (offset 1). We prove this conjecture (Theorem 1).

Prior work and status.

- In the current revision of A127752 (#11, February 11, 2021) the statement is still labelled "(conjecture)". On September 29, 2018, Antti Karttunen added the remark that "the conjecture is true at least up to 2048 first terms" (noting that the indexing of A127752 starts at 0), a b-file of $a(n)$ for $n \le 2048$, and a PARI program; Jean-François Alcover

Date: October 25, 2026.

added a Mathematica program on February 11, 2021. Both programs invert a fixed finite truncation of A. The entry has no formula section [10].

- The triangle entry A127751 (revision #5, September 29, 2018) repeats the conjecture: "Row sums A127752 mod 2 are conjectured to be first Feigenbaum symbolic sequence A035263" [9].
- The companion entries A127749 and A127750 (Barry, same day) concern the triangle with $1/(2n+1)$ in place of $1/(3n+1)$ and conjecture that the row sums of its inverse are, modulo 2, the Fredholm–Rueppel sequence A036987 [7, 8]; R. J. Mathar noted in 2021 that this is equivalent to the row-sum conjecture in A111967 [7, 6]. These are still labelled as conjectures; see Remark 7.
- S. Fried [2] proved conjectures of Barry about the triangle A127793, the inverse of the band triangle with the same support $k \le n \le 2k$ but entries $1/\lfloor (n+2)/2 \rfloor$, and its connection with the largest odd divisor A000265. That paper does not treat A127752.

The math0 referees searched the OEIS (including all entries that cite A127751 or A127752), arXiv, the web and recent papers proving OEIS conjectures, and found no earlier proof or counterexample; the statement is not in the formal-conjectures repository [1]. We therefore classify Theorem 1 as a *new result*. The proof is short and elementary; the only point is to compare consecutive equations of the triangular system $A\,a = \mathbf{1}$.

2. Statement

Write $\mathbf{d}(n) = \text{A035263}(n)$ for $n \ge 1$. Let σ be the morphism of the free monoid on $\{0,1\}$ with $\sigma(0) = 11$ and $\sigma(1) = 10$, and let $w_k = \sigma^k(1)$, so $w_0 = 1$, $w_1 = 10$, $w_2 = 1011$, $w_3 = 10111010$. Each w_k has length 2^k and is a prefix of w_{k+1} (by induction, since w_0 is a prefix of w_1 and σ preserves prefixes), so the w_k converge to an infinite word $\mathbf{d}(1)\mathbf{d}(2)\mathbf{d}(3)\cdots$.

Theorem 1 (Barry's conjecture)**.** *For every $n \ge 0$,*

$$a(n) \equiv \mathbf{d}(n+1) \pmod 2.$$

Equivalently, $a(n)$ is odd if and only if $\mathbf{d}(n+1) = 1$.

The conjecture and its indexing. The comment quoted in the introduction is unsigned; as is usual in the OEIS, we attribute it to the author of the entry, Paul Barry (January 28, 2007). A127752 has offset 0 and A035263 has offset 1, and the conjecture compares the two sequences term by term from their first terms: $a(0), a(1), a(2), \ldots$ modulo 2 is $1, 0, 1, 1, 1, 0, \cdots = \mathbf{d}(1), \mathbf{d}(2), \mathbf{d}(3), \ldots$. This is the reading of Karttunen's remark in the entry ("please note the different indexing, here 0-based"); the other alignment, $a(n) \equiv \mathbf{d}(n)$, fails already at $n = 1$, where $a(1) = 4$ and $\mathbf{d}(1) = 1$.

Lean formalization. The statement was formalized by math0 in `lean/Math0/Statements/OEIS/A127752.lean`, reviewed for fidelity, and frozen (SHA-256 `9f79e91f...0f38ec`) before any proof attempt. Its definitions and the statement read as follows (verbatim and in order; the copyright header, the import, comments, docstrings and all theorems are omitted):

```
namespace Math0.Statements.A127752
open Finset Matrix
def A (n k : ℕ) : ℚ := if k ≤ n ∧ n ≤ 2 * k then 1 / (3 * n + 1 : ℚ) else 0
def next (l : List ℤ) : ℤ :=
  3 * l.length + 1 - ∑ j ∈ range l.length, if l.length ≤ 2 * j then l.getD j 0 else 0
def vals : ℕ → List ℤ
  | 0 => []
  | n + 1 => vals n ++ [next (vals n)]
def a (n : ℕ) : ℤ := (vals (n + 1)).getD n 0
```

```
def Amat (N : ℕ) : Matrix (Fin N) (Fin N) ℚ := Matrix.of fun i j => A i j
def sigma (x : ℕ) : List ℕ := if x = 0 then [1, 1] else [1, 0]
def word : ℕ → List ℕ
  | 0 => [1]
  | k + 1 => (word k).flatMap sigma
def A035263 (n : ℕ) : ℕ := (word n).getD (n - 1) 0
def statement : Prop := ∀ n : ℕ, a n % 2 = (A035263 (n + 1) : ℤ)
end Math0.Statements.A127752
```

The triangle and its row sums. `A n k` is $A(n,k)$, a rational number, and `Amat N` is the leading $N \times N$ block A_N of A. The sequence a is not defined through a matrix inverse but by forward substitution in $A\,s = \mathbf{1}$: row n of this system reads $\frac{1}{3n+1}\sum_{\lceil n/2\rceil \le j \le n} s(j) = 1$, so

$$s(n) = 3n + 1 - \sum_{\substack{j<n \\ n\le 2j}} s(j). \tag{1}$$

The list `vals n` is $[s(0), \dots, s(n-1)]$: `vals (n+1)` appends to `vals n` the value `next (vals n)`, which is $s(n)$ computed by (1) from the entries already in the list (so each value is computed once). Then `a n` is the entry of `vals (n+1)` at the 0-based position n, that is, $s(n)$. The statement file proves (statements shown, proofs omitted)

```
theorem a_rec (n : ℕ) :
    a n = 3 * n + 1 - ∑ j ∈ range n, if n ≤ 2 * j then a j else 0
theorem a_spec (N : ℕ) (i : Fin N) : (a i : ℚ) = ∑ j, (Amat N)⁻¹ i j
```

Thus `a_rec` is (1), and `a_spec` ties `a` to the OEIS definition: for all $i < N$, $a(i)$ is the i-th row sum of Mathlib's inverse A_N^{-1}. (Its proof: A_N is lower triangular with determinant $\prod_{n<N}(3n+1)^{-1} \neq 0$, and $A_N(a(0), \dots, a(N-1))^T = \mathbf{1}$ row by row by (1).) As the leading blocks of A^{-1} are the A_N^{-1}, this is the i-th row sum of A^{-1}. In particular $a(n)$ is an integer, and "$a(n)$ mod 2" is encoded as `a n % 2`, the Euclidean remainder in $\mathbb{Z}$, which is 0 or 1.

The sequence A035263. `sigma` is σ (on all of $\mathbb{N}$, with every nonzero letter treated as 1; only the letters 0 and 1 occur), `word k` is w_k as a list, and `A035263 n` is the letter of w_n at the 1-based position n, which exists since $|w_n| = 2^n \ge n$. The statement file proves that the choice of the iterate does not matter:

```
theorem word_length (k : ℕ) : (word k).length = 2 ^ k
theorem word_prefix (k : ℕ) : word k <+: word (k + 1)
theorem A035263_eq {n k : ℕ} (hn : 1 ≤ n) (hk : n ≤ 2 ^ k) :
    A035263 n = (word k).getD (n - 1) 0
```

(`<+:` is the prefix relation; `l.getD i 0` is the entry of `l` at the 0-based position i, or 0 if i is not less than the length of `l`.)

Data checks. The theorems `a_data` and `A035263_data`, proved by `decide +kernel`, state that the definitions reproduce all 106 terms $a(0), \dots, a(105)$ listed in A127752 and all 105 terms $\mathbf{d}(1), \dots, \mathbf{d}(105)$ listed in A035263. This anchors the definitions to the OEIS independently of the conjecture.

The statement. `statement` is $\forall n \ge 0,\ a(n) \bmod 2 = \mathbf{d}(n+1)$, which is Theorem 1 with the indexing discussed above. The statement file uses no `sorry`, `native_decide` or custom axioms.

3. Proof

The proof follows the Lean file `lean/Math0/Frontier/OEIS/A127752.lean`; the Lean name of each step is given in brackets. All lemmas are about the sequence a defined by (1), which by `a_spec` is A127752.

3.1. **A recursion for a.**

Lemma 2 (window identity; `sum_Ico_a`)**.** *For every $n \geq 0$,*

$$\sum_{j=\lfloor (n+1)/2 \rfloor}^{n} a(j) = 3n + 1.$$

Proof. For $0 \leq j \leq n$ we have $n \leq 2j$ if and only if $j \geq \lceil n/2 \rceil = \lfloor (n+1)/2 \rfloor$, and $j = n$ satisfies this. Hence the sum in (1) together with the term $a(n)$ is the sum over $\lfloor (n+1)/2 \rfloor \leq j \leq n$, and (1) (`a_rec`) rearranges to the claim. In Lean the index set is identified with `Ico` $\lfloor (n+1)/2 \rfloor$ $(n+1)$ by `omega`. □

This is row n of the system $A\,a = \mathbf{1}$ multiplied by $3n + 1$.

Lemma 3 (`a_zero`, `a_even`, `a_odd`)**.** *For every $m \geq 0$,*

$$a(0) = 1, \qquad a(2m+2) = 3, \qquad a(2m+1) = a(m) + 3.$$

Proof. For $n = 0$ the sum in (1) is empty, so $a(0) = 1$. Lemma 2 at $n = 2m$, $2m+1$ and $2m+2$ gives, since $\lfloor (2m+1)/2 \rfloor = m$ and $\lfloor (2m+2)/2 \rfloor = \lfloor (2m+3)/2 \rfloor = m+1$,

$$a(m) + \sum_{j=m+1}^{2m} a(j) = 6m + 1, \tag{2}$$

$$\sum_{j=m+1}^{2m} a(j) + a(2m+1) = 6m + 4, \tag{3}$$

$$\sum_{j=m+1}^{2m+1} a(j) + a(2m+2) = 6m + 7, \tag{4}$$

where in (2) the first term of the window was split off and in (3), (4) the last term (for $m = 0$ the sums over $m+1 \leq j \leq 2m$ are empty). The left-hand side of (3) is $\sum_{j=m+1}^{2m+1} a(j)$, so subtracting (3) from (4) gives $a(2m+2) = 3$, and subtracting (2) from (3) gives $a(2m+1) - a(m) = 3$. In Lean the subtractions are done by `linarith`. □

In particular $a(n)$ is odd for every even n (it is 1 or 3), and for odd $n = 2m+1$ the parity of $a(n)$ is opposite to that of $a(m)$.

3.2. **A recursion for A035263.** For a finite word $u = u_0 u_1 \cdots u_{L-1}$ over $\{0, 1\}$ (indexed from 0), $\sigma(u) = \sigma(u_0)\sigma(u_1)\cdots\sigma(u_{L-1})$ is a concatenation of blocks of length 2, and $\sigma(x) = 1\,\overline{x}$ with $\overline{0} = 1$, $\overline{1} = 0$ (`sigma_eq`). Reading off positions gives:

Lemma 4 (`flatMap_getD_even`, `flatMap_getD_odd`)**.** *Let u be a finite word of length L and let $0 \leq i < L$. Then the letter of $\sigma(u)$ at the 0-based position $2i$ is 1, and the letter at position $2i+1$ is 1 if $u_i = 0$ and 0 otherwise.*

Proof. By induction on u. If $u = xu'$, then $\sigma(u) = 1\,\overline{x}\,\sigma(u')$. For $i = 0$ the letters at positions 0 and 1 are 1 and $\overline{x}$. For $i \geq 1$, positions $2i$ and $2i+1$ of $\sigma(u)$ are positions $2(i-1)$ and $2(i-1)+1$ of $\sigma(u')$, and $u_i = u'_{i-1}$ with $i - 1 < L - 1 = |u'|$, so the induction hypothesis applies. □

Lemma 5 (`A035263_odd_index`, `A035263_even_index`)**.** *For every $m \geq 0$,*

$$\mathbf{d}(2m+1) = 1, \qquad \mathbf{d}(2m+2) = \begin{cases} 1, & \mathbf{d}(m+1) = 0, \\ 0, & \text{otherwise.} \end{cases}$$

Proof. Since $m+1 \le 2^m$ (`succ_le_two_pow`), we have $2m+2 \le 2^{m+1} = |w_{m+1}|$, so by `A035263_eq` the letters $\mathbf{d}(2m+1)$ and $\mathbf{d}(2m+2)$ are the letters of $w_{m+1} = \sigma(w_m)$ at the 0-based positions $2m$ and $2m+1$. Here $m < 2^m = |w_m|$, so Lemma 4 applies with $u = w_m$ and $i = m$: the first letter is 1, and the second is 1 if the letter of w_m at position m is 0, and 0 otherwise. Finally, again by `A035263_eq` (with $n = m+1 \le 2^m$), the letter of w_m at position m is $\mathbf{d}(m+1)$. □

In other words, $c(m) = \mathbf{d}(m+1)$ satisfies $c(2m) = 1$ and $c(2m+1) = 1 - c(m)$; this is the familiar self-similarity of the period-doubling word, here derived from the definition of A035263 as a morphic word.

3.3. **Proof of the theorem.**

Proof of Theorem 1. We show $a(n) \bmod 2 = \mathbf{d}(n+1)$ by strong induction on n (`result`). Write $n = 2m$ or $n = 2m+1$.

Case $n = 2m$. By Lemma 5, $\mathbf{d}(2m+1) = 1$. By Lemma 3, $a(0) = 1$ and $a(2m) = 3$ for $m \ge 1$, so $a(2m) \bmod 2 = 1$ in both cases.

Case $n = 2m+1$. Since $m < 2m+1$, the induction hypothesis gives $a(m) \bmod 2 = \mathbf{d}(m+1)$. By Lemmas 3 and 5, $a(2m+1) = a(m)+3$ and $\mathbf{d}(2m+2) = 1$ if $\mathbf{d}(m+1) = 0$, and $\mathbf{d}(2m+2) = 0$ otherwise. If $\mathbf{d}(m+1) = 0$, then $a(m)$ is even, so $a(m)+3$ is odd and $a(2m+1) \bmod 2 = 1 = \mathbf{d}(2m+2)$. If $\mathbf{d}(m+1) \neq 0$, then $a(m) \bmod 2 = \mathbf{d}(m+1)$ is a nonzero element of $\{0,1\}$, so $a(m)$ is odd, $a(m)+3$ is even, and $a(2m+1) \bmod 2 = 0 = \mathbf{d}(2m+2)$.

In Lean the case split is `Nat.even_or_odd'`. The even case is closed by `rfl` after rewriting with `a_zero` or `a_even`. In the odd case the two values of the `if` are separated by `split_ifs`, and each parity step is closed by `omega` after moving the casts from $\mathbb{N}$ to $\mathbb{Z}$. □

3.4. **Further remarks (not formalized).**

The following remarks are proved on paper only; they are not part of the Lean development and were not reviewed by the math0 referees.

Remark 6 (closed form)**.** Write $n+1 = 2^t u$ with u odd. Lemma 3 determines a completely: in terms of $b(N) = a(N-1)$ it reads $b(1) = 1$, $b(u) = 3$ for odd $u \ge 3$, and $b(2M) = b(M) + 3$ for $M \ge 1$. Hence

$$a(n) = \begin{cases} 3t+1, & u = 1 \ (n+1 \text{ a power of 2}), \\ 3t+3, & u \ge 3, \end{cases}$$

in agreement with the data ($a(7) = 10$, $a(11) = 9$, $a(63) = 19$). Since $\mathbf{d}(n+1) = 1$ if and only if t is even (by Lemma 5, $\mathbf{d}(N) = 1$ for odd N and $\mathbf{d}(2M) = 1 - \mathbf{d}(M)$; this is the description of A035263 in its name as the parity of the 2-adic valuation of $2N$ [5]), this gives a second proof of Theorem 1: $a(n) \equiv t+1 \pmod 2$ in both cases. The Lean proof does not use this closed form.

Remark 7 (the companion conjectures)**.** The argument of Lemmas 2 and 3 applies to any band triangle with entries $1/\gamma(n)$ for $k \le n \le 2k$ and $\gamma(n) \neq 0$: the row sums s of its inverse satisfy $\sum_{j=\lfloor (n+1)/2 \rfloor}^{n} s(j) = \gamma(n)$, hence $s(0) = \gamma(0)$, $s(2m+2) = \gamma(2m+2) - \gamma(2m+1)$ and $s(2m+1) = s(m) + \gamma(2m+1) - \gamma(2m)$. For $\gamma(n) = 2n+1$ (A127749, A127750) this gives $s(0) = 1$, $s(2m+2) = 2$, $s(2m+1) = s(m)+2$, so $s(n)$ is odd if and only if $n+1$ is a power of 2; that is, $s(n) \bmod 2$ is the Fredholm–Rueppel sequence A036987 (offset 0), as Barry conjectured [7, 8]. For $\gamma \equiv 1$ (the 0/1 band triangle, which is A101688 read by rows; its inverse is A111967) it gives $s(0) = 1$, $s(2m+2) = 0$, $s(2m+1) = s(m)$, so the row sums are exactly A036987, which is the conjecture recorded in A111967 [6].

4. Machine verification

Files and declarations. The math0 repository is `https://github.com/Foundation-0/math0`. The proof is the file `lean/Math0/Frontier/OEIS/A127752.lean`; it imports the frozen statement file `lean/Math0/Statements/OEIS/A127752.lean` (Section 2) and the audit commands in `lean/Math0/Audit.lean`. Its main declaration is the theorem `Math0.Frontier.A127752.result`, whose type is `Math0.Statements.A127752.statement`. The auxiliary lemmas, named in Section 3, are private. The proof of `result` is a strong induction on n (`Nat.strong_induction_on`) with the case split `Nat.even_or_odd'`, exactly as in Section 3. The file ends with the audit command `#math0_settles Math0.Frontier.A127752.result Math0.Statements.A127752.statement`. Neither file uses `sorry`, `native_decide` or custom axioms; the only option set is `maxRecDepth`, for the data check `a_data` in the statement file.

Audit. The command `#math0_settles`, defined in `lean/Math0/Audit.lean` for statements that math0 formalizes itself, fails unless (1) `statement` is a `Prop`-valued definition without universe parameters; (2) `statement` and `result` depend on no axioms besides `propext`, `Classical.choice` and `Quot.sound` (this rejects `sorryAx`, `Lean.ofReduceBool` from `native_decide`, and custom axioms); and (3) the type of `result` is syntactically `statement` or $\neg$ `statement`, up to metadata annotations, with no unfolding. Here it reports

```
math0 audit: 'Math0.Statements.A127752.statement' depends on axioms [Classical.choice, Quot.sound, propext]
math0 audit: 'Math0.Frontier.A127752.result' depends on axioms [Classical.choice, Quot.sound, propext]
math0 statement check: 'Math0.Frontier.A127752.result' proves 'Math0.Statements.A127752.statement'
```

Toolchain and reproduction. We use Lean 4 [4] `v4.33.1` and Mathlib [3] `v4.33.1`, pinned in `lean/`; the repository also pins formal-conjectures [1] at commit `c77fdd4`, which this result does not use. To reproduce, run `scripts/verify.sh` from the repository root: it fetches the Mathlib cache, builds all math0 results with `lake build` (re-running every audit command), and checks that the pinned dependencies are unmodified. The single file can be checked with `lake env lean Math0/Frontier/OEIS/A127752.lean` in the directory `lean`.

References

[1] M. Firsching, P. Lezeau, S. Mercuri, M. Z. Horváth, Y. Dillies, C. Sönne, E. Wieser, F. Zhang, T. Hubert, B. Agüera y Arcas, and P. Kohli, *Formal Conjectures: An Open and Evolving Benchmark for Verified Discovery in Mathematics*, arXiv:2605.13171 [cs.AI], 2026. Repository: `https://github.com/google-deepmind/formal-conjectures`.

[2] S. Fried, *On the inverse of a certain triangular matrix and its connection to the largest odd divisor*, Notes on Number Theory and Discrete Mathematics **32**(2) (2026), 255–262, `https://doi.org/10.7546/nntdm.2026.32.2.255-262`.

[3] The mathlib Community, *The Lean mathematical library*, in: Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), ACM, 2020, 367–381.

[4] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in: Automated Deduction – CADE 28, Lecture Notes in Comput. Sci. **12699**, Springer, 2021, 625–635.

[5] The OEIS Foundation Inc., *Trajectory of* 1 *under the morphism* $0 \to 11$, $1 \to 10$*; parity of* 2*-adic valuation of* $2n$*:* $a(n) = \mathrm{A000035}(\mathrm{A001511}(n))$, entry A035263 (K. Karamanos and N. J. A. Sloane) in The On-Line Encyclopedia of Integer Sequences, `https://oeis.org/A035263`.

[6] The OEIS Foundation Inc., *Inverse of number triangle A101688*, entry A111967 (P. Barry, 2005) in The On-Line Encyclopedia of Integer Sequences, `https://oeis.org/A111967`.

[7] The OEIS Foundation Inc., *Inverse of number triangle* $A(n,k) = 1/(2n+1)$ *if* $k \le n \le 2k$, 0 *otherwise*, entry A127749 (P. Barry, 2007) in The On-Line Encyclopedia of Integer Sequences, `https://oeis.org/A127749`.

[8] The OEIS Foundation Inc., *Row sums of inverse of number triangle* $A(n,k) = 1/(2n+1)$ *if* $k \le n \le 2k$, 0 *otherwise*, entry A127750 (P. Barry, 2007) in The On-Line Encyclopedia of Integer Sequences, `https://oeis.org/A127750`.

[9] The OEIS Foundation Inc., *Inverse of number triangle* $A(n,k) = \text{if}(k \leq n, \text{if}(n \leq 2k, 1/(3n+1), 0), 0)$, entry A127751 (P. Barry, 2007) in The On-Line Encyclopedia of Integer Sequences, `https://oeis.org/A127751`; revision #5, September 29, 2018.

[10] The OEIS Foundation Inc., *Row sums of inverse of number triangle* $A(n,k) = 1/(3n+1)$ *if* $k \leq n \leq 2k$, 0 *otherwise*, entry A127752 (P. Barry, 2007) in The On-Line Encyclopedia of Integer Sequences, `https://oeis.org/A127752`; revision #11, February 11, 2021.

Produced by Claude (an AI model developed by Anthropic) in the math0 project